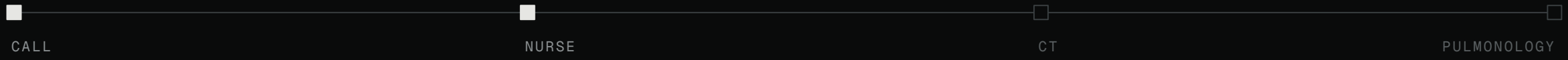
JUNCTION 2026 · ASTRAZENECA: EQUAL CARE ACROSS FINLAND

Polku

Every cancer pathway, on the clock.



Matti is 71.
He has been coughing for seven weeks.

ILLUSTRATIVE PATIENT (SYNTHETIC)

He hasn't called anyone. He doesn't want to be a bother.

> “The enemy isn't cancer. It's silence: waiting without knowing whether anything is happening, and nobody counting.”

SYNTHETIC

Day 49

SINCE FIRST SYMPTOM

NO CONTACT WITH CARE YET

Source: illustrative persona, synthetic. Seven weeks = 49 days.

The gap.
Found late. Waits not counted.

WHAT FINLAND PUBLISHES, AND WHAT IT DOES NOT

65-70 %

of lung cancers in Finland are found at an advanced stage.

REAL, PUBLIC AGGREGATE

Filha lung cancer programme 2026-2035 (citing Knuuttila 2021)

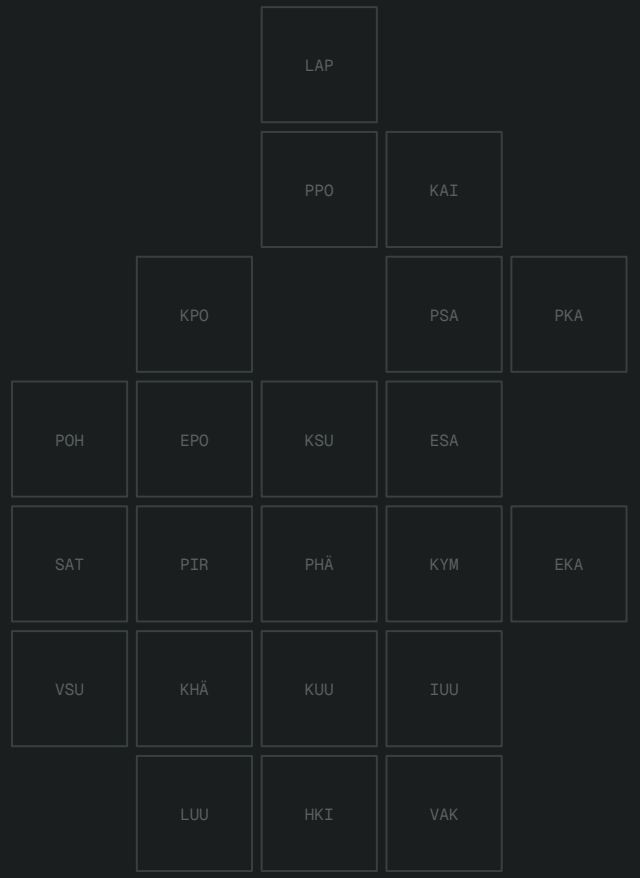
68.9-88.7 %

of referrals to specialised care processed within 3 days, by county, 2025. National 76.8 %. All specialities, not lung-specific.

REAL, PUBLIC AGGREGATE

THL Sotkanet, indicator 13

Lung cancer pathway times per county: not published



We found no public county-level series. Denmark and Sweden publish per region.

Sources: filha.fi lung cancer programme 2026-2035 · sotkanet.fi indicator 13

Care doesn't stall inside the steps. It stalls between them.

LEG 1

First contact to nurse

TODAY

Nobody owns the wait.

WITH POLKU

Nurse callback within 2 working days.

POLKU PROPOSAL, NOT A PUBLISHED TARGET

LEG 2

Referral to CT

TODAY

Published guidance: CT "usually within a week".

WITH POLKU

A clock on that target, visible to the nurse and the next step.

REAL, PUBLISHED GUIDANCE

Pirkanmaa wellbeing county, pirha.fi

LEG 3

CT to pulmonology

TODAY

Published guidance: "goal 2-3 weeks, typically about a month".

WITH POLKU

A clock on that target. At risk is flagged before it breaches.

REAL, PUBLISHED GUIDANCE

Keski-Suomi (HYVAKS), hyvaks.fi

PILOT HYPOTHESIS, TO BE TESTED WITH A BASELINE EXTRACT OF REFERRAL AND PATHWAY TIMESTAMPS

Sources: pirha.fi · hyvaks.fi

The AI listens.
The rules decide. The nurse confirms.

SIX STEPS, ONE SECURED RECORD

EXAMPLE MATCH

New cough for 6 weeks in a former smoker over 40.
Source: Swedish standardised lung cancer pathway v1.4.

REAL, PUBLISHED CRITERION (SE SVF)

01
Caller phones the line.
VOICE AGENT

02
Speech becomes typed symptoms, caller's own words attached.
AI TRANSCRIBES

03
Deterministic rules match a published Nordic referral criterion, with its source.
RULES ENGINE

04
Nurse gets one task with a rule-proposed time: Accept.
NURSE

05
Agent calls the patient back and books it.
VOICE AGENT

06
Same secured record moves to the next step. Every access logged.
AUDIT LOG

Source: Swedish SVF lung cancer v1.4, kunskapsbanken.cancercentrum.se

The most important thing our AI does is nothing.

Five safeguards, enforced in code, not in a prompt.

```
AI SCOPE: SPEECH TO TYPED SYMPTOMS.
NOTHING ELSE.
```

01 **The model only transcribes.**

It turns speech into typed symptoms. It never diagnoses or sets deadlines.

02 **Red flags always mean "Call 112".**

Massive haemoptysis, severe breathlessness, sudden severe chest pain. Enforced by the engine, whatever the model says.

03 **Nothing before identification.**

Three failed codes lock the call. No tool can look up another person.

04 **A nurse makes every clinical decision.**

Polku proposes a time. The nurse accepts it or changes it.

05 **Every record access is logged.**

Each access is recorded as allowed or refused.

Source: Polku system design, as built (demo: polku.eros.ender.fi)

Pathways by 2028, says the strategy.
We built one this weekend.

LIVE DEMO

Covers the call with its handoff packet, the nurse task view, the agent callback, and the handoff strip with its access log.

Demo polku.eros.ender.fi Code github.com/3nd3r1/polku (MIT)

PATHWAY AS CODE

Lung is one typed, cited file. The next cancer is the next file.

STACK, STATED HONESTLY

ElevenLabs voice agent and phone line
Qwen3 on Verda GPUs in the EU as the language model
Deterministic rules engine in TypeScript, with tests
Postgres on Aiven: ontology, cases, clocks and audit

FOUR DEMO SCREENS

01
Call
Symptoms fill a handoff packet, next to the caller's own words.

02
Nurse task
One task with the matched criterion and a rule-proposed time: Accept.

03
Callback
The agent calls the patient back and books the slot.

04
Handoff strip
One line per patient: who has them, the deadline, the access log.

Source: national cancer strategy roadmap 2026-2031, measure 3.1.2 (pathways by 2028)

Three tech partners.
Each one has one job.

WHERE THE CALL GOES

01 VOICE

ElevenLabs

The phone line.

The voice agent answers the call, identifies the caller and asks what the nurse needs. After the nurse accepts, it calls the patient back and books the time.

VOICE AGENT · PHONE NUMBER

02 LANGUAGE MODEL

Verda

Qwen3, hosted in the EU.

Qwen3 runs on Verda GPUs in an EU data centre, on infrastructure we control. It turns the caller's speech into typed symptoms. It never diagnoses.

QWEN3-30B · vLLM · EU

03 RECORD

Aiven

Postgres with the ontology.

Managed Postgres holds the ontology: one secured record per patient, the cases, the step clocks, and a log of every access.

POSTGRES · ONTOLOGY · AUDIT LOG

1. Caller phones the line.
   ELEVENLABS
2. Speech becomes typed symptoms.
   QWEN3 ON VERDA
3. Rules match a cited criterion.
   RULES ENGINE
4. Record, clock and log saved.
   AIVEN POSTGRES
5. Nurse accepts the time.
   NURSE
6. Agent calls back and books.
   ELEVENLABS

Source: Polku system design, as built, github.com/3nd3r1/polku

Built for the measures the strategy already names.

NATIONAL CANCER STRATEGY 2026-2035 AND ROADMAP 2026-2031

MEASURE 2.5.2

Symptom-based referral criteria.

MEASURE 3.5.3

Digital pathways.

MEASURE 3.1.2

Pathways with target access times.

GOAL 3 INDICATOR

Share of patients starting treatment within a county-agreed target.

STRATEGY TARGET, 5-YEAR LUNG CANCER SURVIVAL

14+ points

The strategy aims to raise 5-year lung cancer survival by more than 14 points, to the Swedish and Norwegian average (p.5).

REAL, PUBLIC TARGET

National cancer strategy 2026-2035, p.5

REAL, PUBLIC STRATEGY TEXT

Roadmap pp.52-57, 73-74

Source: national cancer strategy 2026-2035 p.5; roadmap 2026-2031 pp.52-57, 73-74

Approve a 90-day pilot in one wellbeing county.

THREE PROPOSED LOCAL TARGETS

2

WORKING DAYS

Nurse callback within 2 working days.

POLKU PROPOSAL, NOT A PUBLISHED TARGET

7

DAYS

CT within about a week of a confirmed referral.

DERIVED FROM COUNTY GUIDANCE

Pirkanmaa: "usually within a week", pirha.fi

21

DAYS

First pulmonology visit within 2-3 weeks.

DERIVED FROM COUNTY GUIDANCE

Keski-Suomi (HYVAKS): "goal 2-3 weeks, typically about a month", hyvaks.fi

WHAT IT TAKES (PLANNING ASSUMPTIONS)

One phone number, one nurse workflow, one pulmonology clinic, a clinical sponsor, a baseline extract of referral and pathway timestamps, about 5 nurse hours a week, 2 protected CT slots a week. No saving is forecast.

Introduce us to one wellbeing county.

Sources: pirha.fi · hyvaks.fi · targets are pilot proposals

Honesty notes.

What is real, what is synthetic, and what is not built yet.

SYNTHETIC	Patients, identity codes and pathway clocks in the demo. Mobile ID is simulated.
REAL	Filha, THL Sotkanet and strategy figures are public aggregates, each shown with its source.
NOT YET	The voice layer runs on ElevenLabs; EU-only processing is a design goal, not the current state. The callback rings in the browser. A research prototype, not for clinical use; it needs a regulatory assessment.

RESEARCH PROTOTYPE, NOT FOR CLINICAL USE

Every cancer pathway, on the clock.

DEMO AND CODE

polku.eros.ender.fi github.com/3nd3r1/polku (MIT)